KONSTANTIN GRIGOROV

PRINCIPAL ENGINEER DATA AND AI PLATFORMS

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PROFILE

Principal data and AI engineer with 10+ years building trading, investment and enterprise platforms. Combines hands-on architecture and software delivery across Databricks, Azure, AWS, Kubernetes and Terraform with front-office domain knowledge and technical leadership. At Shell Trading, built reusable platform capabilities that let engineering teams deliver governed data pipelines and AI-enabled data products faster.

LEADERSHIP PROFILE

Platform architecture Lakehouse and data product strategy, Databricks, Delta Lake, Unity Catalog, PySpark and Polars
Engineering enablement Reusable pipeline frameworks, platform standards, CI/CD, testing, observability and secure delivery
Cloud and operations AWS and Azure, Kubernetes, Docker, Terraform, infrastructure as code and event-driven systems
AI and domain MCP, semantic retrieval, agent evaluation, trading platforms, pricing and market data

PROFESSIONAL EXPERIENCE

Shell Trading and Supply — Jun 2022 - Present

London, United Kingdom

Principal Engineer Data and AI in Trading | Principal+ — 2026 - Present
Senior Technical Lead | Principal Level — Jun 2024 - Sep 2026
Senior Platform Engineer | Front Office — Jun 2022 - May 2024

Hands-on technical lead for trading data products, reusable platform capabilities and applied AI, working directly with desk users and enterprise engineering teams.

- Architected and built a reusable Python and Polars data engineering framework that enables engineers to develop, test and operate production pipelines across landing, raw, staging and Delta layers. The framework was proven on workloads exceeding 10 million rows and helped reduce pipeline delivery from two months to three weeks.
- Established shared patterns for stage orchestration, incremental change processing, audit metadata, data quality, observability, testing and deployment across a 16-engineer data community.
- Led engineering of a flagship product in the Chief Data Officer portfolio, co-architecting a cross-desk data catalogue spanning natural gas, crude and middle distillates.
- Built multiple data and AI products from inception for a 10,000+ employee trading organisation, with solutions recognised by enterprise architecture.
- Led architecture assessment and roadmap work for shared data-processing capabilities, balancing platform reuse, migration risk, event-driven orchestration and production readiness across multiple data products.
- Shaped MCP-based data discovery and agent architecture, including governed tool routing, semantic retrieval, authentication boundaries, reusable skills and evaluation patterns.
- Built one of Shell's first Excel and ChatGPT integrations, presented it to senior leadership, and led a global AI hackathon for nearly 100 participants across two continents.

PROFESSIONAL EXPERIENCE CONTINUED

Senior Software Engineer | Trading 212 — May 2021 - Apr 2022

London, United Kingdom

Built pricing, market-data and risk automation for a high-volume retail trading platform in close partnership with front-office users.

- Re-engineered core pricing flows to improve quote competitiveness and accelerate market-data processing.
- Designed a Python algorithmic trading engine that automated equity position management across 17,000+ instruments.
- Improved authentication services supporting 40,000+ daily active users and delivered extensible market-data and reporting frameworks using Interactive Brokers APIs.

Senior Associate Cloud Native Software Engineer | J.P. Morgan — Jan 2020 - May 2021

London, United Kingdom

Modernised cloud-native infrastructure and trading workloads across a global engineering organisation.

- Built an enterprise automation library for controlled Terraform module updates across global engineering teams.
- Developed a Kubernetes operator for consistent orchestration of platform-critical schedulers.
- Migrated risk and trading workloads to AWS using infrastructure as code and Kubernetes.
- Contributed to Athena, J.P. Morgan's flagship cross-asset trading and risk platform.

Software Engineer, Algorithms | Playtech — Feb 2018 - Jan 2020

London, United Kingdom

- Built FastAPI microservices within enterprise CI/CD standards and modernised Kubernetes deployments and reliability automation.
- Improved sports-pricing models, including a 20% performance improvement for core baseball algorithms.

Full-Stack Developer, Lottery | Playtech — Jul 2017 - Feb 2018

London, United Kingdom

- Built Python and OpenCV barcode-detection automation and delivered messaging and reporting features across React and Oracle systems.

Software Engineer Graduate Scheme | The Hut Group — Sep 2016 - May 2017

Manchester, United Kingdom

- Developed Java and SQL services supporting logistics operations and delivered changes through Jenkins-based continuous integration.

TECHNOLOGY

Python, Go, SQL, TypeScript, PySpark, Polars, Databricks, Delta Lake, Unity Catalog, Azure, AWS, GCP, Kubernetes, Docker, Terraform, GitHub Actions, Azure DevOps, FastAPI, Airflow, MLflow, Power BI, MCP, LangChain, vector databases, semantic retrieval, agent evaluation and distributed systems

EDUCATION

MEng Computer Science | The University of Manchester — 2011 - 2016

High 2.1 | Manchester, United Kingdom

LEADERSHIP AND RECOGNITION

- Selected for Shell's Principal+ engineering community after completing a technical defence before a senior engineering panel.
- Nominated for the CIO Innovation Excellence Award 2026 for AI innovation.
- Selected to speak at AI and Big Data Expo Europe 2026 on moving MCP systems from prototype to production.
- Shell Significant Contribution Award for restoring a critical renewables dashboard and reducing expected downtime from one month to three days.